

General comments (last updated 24.03.23)

I will answer some pollev.com/cabe questions in this Google Doc. The more specific the question is, and the more votes it gets, the more attention I can give it. I will "hide" questions if I think I've answered it, or if it was not clear or narrow enough to enable an answer. Feel free to try to make it more specific.

Further below I've inserted some typical questions from last year, and my replies to them.

Last answer

Question: Svaghederne ved cross-sectional studier kan vurderes via AMJ, hvordan kan svaghederne ved longitudinal, eksperiment og casestudie vurderes? /

Weaknesses of cross-sectional studies can be assessed via AMJ insights, how can weaknesses of longitudinal, experiments, case studies be assessed.

Answer: Kriterierne for AMJ kan også bruges på f.eks. longitudinale forskningsdesign. Omitted variable bias og kausalitet kan stadig være en udfordring. Mere generelt vil jeg henvise til en af de første slides i studiecafe 2 (selection bias eller confounding variables f.eks.), hvor jeg giver et overblik over hvad man kan overveje ved forskningsdesign der ikke er eksperimenter. Svagheder ved eksperimenter er; er det vitterligt et eksperiment, samt hvad er ekstern validitets truslerne.

Criteria for AMJ can also be applied on e.g. longitudinal designs. Omitted variable bias and causality can still be a challenge, for example. More generally I'd refer to one of the first slides in studycafe 2 (selection bias or confounding variables e.g.), where I provide an overview of how what to consider when assessing a research design that is not an experiment. The weaknesses of an experiment: is it truly an experiment, and what are the threats to external validity.

Question: På lektion 3 om eksperimenter kommer du ind på en type der hedder "Survey experiment", og referere til det som et "DIFFERENT TYPE OF EXPERIMENT" (slide 20). Jeg kan ikke finde det i pensum teksterne, er dette også noget vi skal kunne? Synes ikke er så fyldestgørende på slides, findes det i en tekst? / Do I need to know more, something specific about survey experiments?

Answer: I just illustrated survey experiments to highlight that an experiment can be executed in various ways and that survey experiments can be easy (or easier) to execute. You don't have to know more about them than this fact (that you can randomize questions to a respondents).

Question: Er abduction en "rodet" proces mellem induktion og deduktion som går frem og tilbage mellem dem, eller er det hvor man eksempelvis starter med deduktiv tilgang og bevæger sig over i induktiv? / Is abduction a "messy" process going back and forth between induction and deduction, or is it a proces where one begins with e.g. deduction and then transitions to induction (and stays there)?

Answer: Abduction is not a kind of process where one begins with e.g. deduction and then transitions to induction and stays there. It is a back-and-forth [=frem og tilbage] process, where one both pays attention to theory and data, in order to find the best possible explanation.

Question: Må vi bruge ordbogen.com? eller google oversæt? / Can we use ordbogen.com or Google Translate?

Answer: No, accessing the internet via a browser is not allowed.

Question: Skal vi vurdere validitet (intern og ekstern) ved andre forskningsdesign end eksperimenter? / Should we evaluate validity (internal and external) in other research designs than experiments?

Answer: Yes, in order to assess the strengths and weaknesses of, for example, a cross-sectional or longitudinal research design it is relevant to talk about validities.

Question: How can we know that one paradigm is superior to another if objective natural truth does not exist? Philosophically, if there is no objective truth, there is no point of comparison.

Answer: Guba claims that it is "an act of faith" to believe one or the other paradigm. There are no objective criteria that can make one state one is more true per se or not. In some contexts, some assumptions make more sense than in others, though.

Question: Vil der i et spørgsmål til et forskningsdesign være tilfælde hvor vi skal komme frem til, at der er brugt flere forskellige designs? / Could there be a question about a research design, where one should conclude that multiple designs were used?

Answer: Yes, that could be the case. We have heard about a few examples in class that involved multiple research designs, e.g. a field experiment that was part of a case study.

Question: Kan du forklare forskellen på den måde Kuhn og Guba bruger paradigmer? / Can you explain how Kuhn's and Guba's use of the term paradigm is different?

Answer: Kuhn uses the word paradigm to refer to the beliefs of a given scientific paradigm. More specifically, the scientific results one believes to be true, the specific methods and techniques used to collect data, what data is considered valuable, what specific research questions are considered meaningful, etc. In economics, it would e.g. be the belief that one can make supply-demand curves, and that these help explain economic dynamics, how one collects data on this, how one would validate / falsify beliefs etc. Guba uses the word paradigm to explain the basic assumptions and beliefs that guide scientific inquiry [=undersøgelser], more specifically ontological, epistemological and methodological assumptions. Guba's approach is thus more abstract, and less focused on specific beliefs that a specific scientific community has. Guba's categorization fits all sciences, while Kuhn thinks that different paradigms in different scientific fields are, different. (and in case anyone is curious about chatbots, ChatGPT-4's answer to this question was not very insightful).

Question: Can we reference the slides when using some quotes in the exam paper? I've seen such practice in the previous exam papers. Does that still hold?

Answer: Yes, you can reference / quote slides.

Question: Vil du forklare ecological validitet (hedder det økologisk?). Hvordan adskiller det sig fra ekstern validitet? Synes ikke der er blevet lagt lige så stort fokus på netop dettes i pensum, men er det en lige så stor del som intern/ekstern? / Can you explain ecological validity? How is ecological validity different from external validity?

Answer: Ecological validity and external validity are related concepts in research, but they refer to different aspects of generalizability in business administration. Ecological validity refers to the extent to which the research findings can be generalized or applied to real-world business settings. It is concerned with the extent to which the research reflects the natural complexity of the environment in which the phenomenon being studied occurs. External validity, on the other hand, refers to the degree to which research findings can be generalized beyond the specific sample and context of the study. It is the extent to which the research findings hold true for other populations, settings, and times.

Hence, a lab study can be externally valid in the sense that it can be generalized to other lab studies, many different populations, and different kinds of tasks in lab studies. It might, however, not have ecological validity in the sense that it might not generalize to a more natural, business environment, in real life. (answer inspired by ChatGPT)

Question: I eksamensopgaven for 2020 kan man vælge at svare på 3a eller 3b. Begge eksempler vælger 3b. Har du et godt eksempel eller nogle ting man kunne komme ind på, hvis man vælger 3a? / In the 2020 exam, one could answer either 3a or 3b. Do you have an example of what to do if one picks 3a?

Answer: The curriculum was slightly different in 2020, and there was a lesson on qualitative vs. quantitative methods. Hence, such a question would not be asked this year.

Question: Nogle af gode besvarelser på eksamensopgaver du har uploadet har en tendens til at svare på andre ting end der bliver spurgt om i spørgsmålet. Skal det forstås som om at det er godt at komme ind på mange ting, selvom det er begrænset hvor meget det relaterer sig til spørgsmålet? / Some of the good uploaded answers to exams have a tendency to respond to many other things than what is being asked in the question. Does that mean that it is good to cover other things, even though it is limited to what is being asked?

Answer: Briefly put, no it is not necessary or desirable as such to cover other things. Some good answers have had the time and ability to explain related concepts in detail. But, you can get the best possible assessment of a given question just based on answering the main elements. If a question asks where Kuhn fits into Guba's paradigms, you could define what positivism is, but since he obviously is not a positivist, this does not make a big difference. The main elements one should cover is what is post-positivism, what is constructivism and what is Kuhn's position (and of course, also what your final answer to the question is). If you have additional elements to cover, you can add them, but it is not something you as such should worry about. An average length exam answer can and often is something that is a top grade.

Question: Hvori består at Kahneman taler om biases, når hans bog kun omhandler system 1 og 2? / Why do we focus on biases in relation to Kahneman, when his book is just about system 1 / 2?

Answer: The below quote from page 25 illustrates why people usually associate the idea of biases with Kahneman: because of how system 1 and 2 operates, we have biases, i.e. systematic errors.

most of the time because System 1 is generally very good at what it does: its models of familiar situations are accurate, its short-term predictions are usually accurate as well, and its initial reactions to challenges are swift and generally appropriate. System 1 has biases, however, systematic errors that it is prone to make in specified circumstances. As we shall see, it sometimes answers

Kahneman 2011: p. 25

Question: Når man kigger på eksperimenter (RCT), er det så et krav, at der SKAL være én kontrolgruppe og én treatmentgruppe. Eller kan det også være et RCT, selvom der er tre gruppe, som får forskellige treatment? Ligesom ordinær eksamen 2021 spg 1 / Do RCTs have to have ONE control group and ONE treatment group?

Answer: Den korte udgave af svaret: Nej. Klassiske hovedpine pille eks. består jo af hovedpine pille, placebo, og ingen pille. Det primære er at der kun er én forskel på tværs af grupperne.

English version of answer, slightly longer: No. The classical headache pill example actually consists of 3 groups; actual headache pill, placebo pill, no pill. Similarly, one can imagine a setup where one is investigating the impact of incentives, and tests the impact of 1 DKK pr. task, 5 DKK pr. task, 10 DKK pr. task (etc.). The main aim is to have ONE difference across conditions. Comparing 1 vs. 5 or 1 vs. 10 etc. all imply one difference (size of the incentive). I understand why one gets this impression, that there can only be one of each, but it is not the case. Also notice that it is not straightforward to identify what the control group is - could be argued to be the one with lowest amount, but not always obvious. Key thing: manipulating only one feature.

Question: ift AMJ artiklen, når vi skal forklare fc. common methods variance skal vi så indsætte citater, eller er det fint bare at forklare betydningen af fænomenet? /

Considering the AMJ article, if we are to explain common methods variance, do we have to add quotes?

Answer: Generally one does not expect quotes or references for technical terminology (such as confounding variables, or common methods variance). You are welcome to use a quote, but typically not expected.

Terms like ontology, epistemology, internal validity etc. would be considered in the same category.

You are expected to quote the text book when a) the question requests it, and b) when it is appropriate and helps you make your point. Quoting / Referring to the paper is a useful signal to send, that says "look, I know this is a good quote, and I thus show good understanding of the text".

Question: Skal vi kunne redegøre for Comparative forskningsdesign, der er nævnt på sider 69-71 i BRM? / Should we have to account for Comparative Research Designs?
Answer: No.

Question: Should we still evaluate whether a research is fulfilling the RCT standard, even if it is not explicitly asked in the question? For example if the question says "what research design do we see here?"

Answer: Only if it is / you suspect it is a RCT design. For example, if it is (clearly) is a case study, or you are not sure if it is a cross-sectional or longitudinal design, then you do not have to introduce RCT standards.

Question: Hvordan skal man citere eks. PoS og BRM med "Havard referencing" når forfatteren og årstal ikke er angivet på disse tekster? / Should we reference using a particular style.

Answer: You do not have to use a particular style when referencing. Author/Year: Page number is my suggestion (Nelson, 2016: p. 1698), but as long as it is clear and consistent what you do, you will be fine. More information can be found in the "reference list" document uploaded to Brightspace (material folder).

Question: Could you maybe summarize the main points of the Holst-Hansen & Bergenholtz 2020 paper in connection to this curriculum? I had a hard time to do so (in comparison to Kuhn or Guba texts...)

Answer: The main purpose with the text is to show you, in detail, how a RCT can be presented in a scientific paper. What information is provided, what does randomization look like in practice, what external validity considerations are provided, etc. So, merely an illustration of what such studies look like.

Question: Vil du forklare igen, hvad Kuhn vs. Aschwanden mener, at ordet "Truth" betyder/dækker over? / Would you explain again, what Kuhn vs. Aschwanden mean with the word "Truth"?

Answer: I provide this quote in lesson 11: "But the scientists have not realized that Kuhn's claim is based on a philosopher's definition of "truth," which is not achieved (if ever) until there is exact knowledge of the ultimate constituents of matter on the quantum-gravity scale. Scientists use a less demanding definition for the word "truth," in which measured parameters can be subject to nonzero error bars." Kenneth G. Wilson, in Physics Today 54, 3, 53 (2001)". Aschwanden has a different definition / use of the word truth. Something that is approximately true, will be considered true (or close enough). Kuhn, implicitly, claims that for something to be true, it has to be exact and ultimate. So, not just close enough. They thus have a different standard for what truth means. If Aschwanden also demanded that only something that was finally and absolutely true could be called true, she might also agree that truth cannot be achieved; it is only temporary and approximate [=tilnærmelsesvis].

Question: Can a field experiment be a RCT / Kan et felt-eksperiment være et RCT

Answer: Yes. The definition of an experiment (as used in this course) is = randomized controlled trial. So, they are equivalent. A field experiment is thus a RCT, carried out in the field, e.g. in an organization.

Question: Can we use other sources than the texts in the curriculum? Kan vi bruge andre kilder end tekster fra pensum?

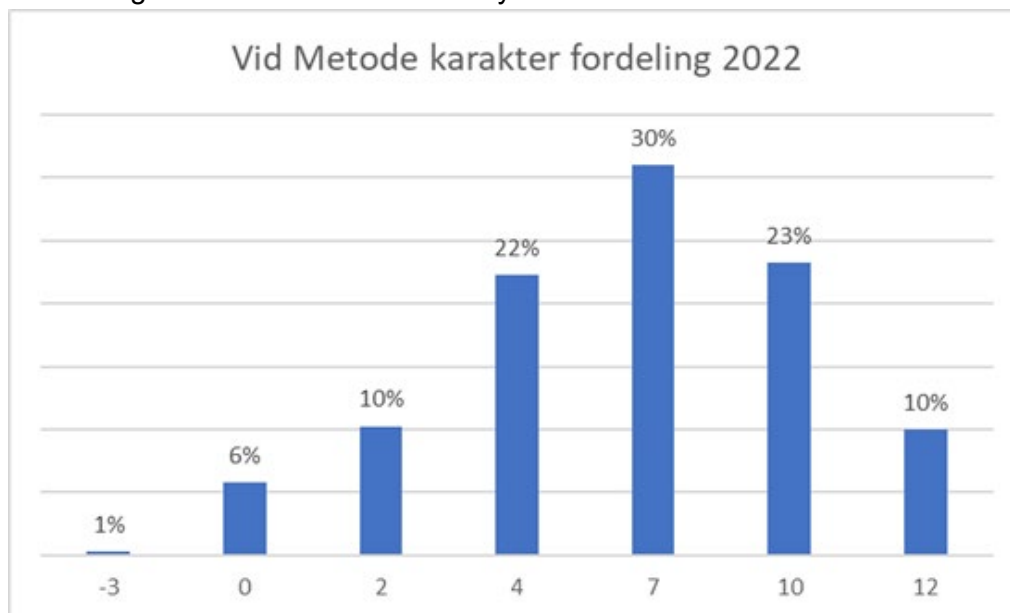
Answer: Yes. Remember to refer to them if you cite them / use ideas from there.

Question: Kuhn's text is more difficult than "PoS chapter 5", which describes Kuhn's ideas. Is it more right/relevant to cite Kuhn than the other one? / Kuhn's tekst er sværere end "PoS 5". Er det mere rigtigt/relevant at citere Kuhn end den anden?

Answer: It is perfectly alright to cite PoS chapter 5. At times Kuhn's text is more precise and / or has more detail than PoS 5, which can make it more useful. But there is nothing wrong in relying on PoS 5.

Question: What is the grade distribution?

Answer: Grade distribution from 2022 (Danish HA exams, roughly similar for BSc). Neither the average nor the distribution usually varies much.



Question: How to prepare for the exam. Read all articles again? Write down definitions? Do you have a checklist for a good answer?

Answer: I've inserted a few slides below (the first one from lesson 1). A few further comments as well in the following. Some of the articles are fairly difficult, so having to read a text twice should not be considered unusual [=usædvanligt]. Having some prepared [=forberedte] notes (e.g. including key quotes) for some of the texts is probably helpful - you might check out exams from 2020-2022 and note that some topics often appear [=optræder ofte]. Generally, you can also consider if there are texts that have been mentioned often throughout the course.

A checklist for a good answer I don't have as such. My best recommendation is to practice former exam questions (e.g. spend at least 15-30 minutes considering what you would write) and then compare with the uploaded good answers.

A FEW KEY POINTS FOR HOW TO PREPARE FOR POS EXAMS

- Use reading guides to read texts. Often read twice.
- Use lectures/slides as pointers to identify key elements and get overview of curriculum
- Create own overview, e.g. including key quotes of key texts
- Read prior exam questions + answer exam answers to test yourself.
- Don't only review notes, but practice exam questions



5 ECTS course ≈ 140 hours
Exam + preparation: 40 hours
 $100 / 15 = 2$ hour lesson + 5 hour preparation
= 7 hours a week

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AARHUS UNIVERSITY
SCHOOL OF BUSINESS AND SOCIAL SCIENCES

A few comments about the exam

- Suggestion for preparation
 - Read curriculum + reflect on exercises/questions in class + study previous exams
 - You'll be too busy to read the curriculum in the exam situation
 - Practice Research Design challenges + look at former exam questions and consider how you would approach them. This year's question will be similar in style
 - This course doesn't call for a template approach, as some other courses.
 - Not a course where one (only) learns a model or table or technique: critical thinking and reflection on what science and truth is about is needed

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Answers to common Pollev questions from prior years

A few questions concerning the exam, and referencing

Question: Is it a requirement to add a reference list at the end of your exam answer?

Answer: This used to be a demand, but the regulations have just changed. For an exam of this length (less than 24 hours), you do not have to provide a reference list for any text that is part of the curriculum. However, any texts that are not in the curriculum, you need to provide a reference for in the reference list. Furthermore, you still have to provide the relevant details

if you quote someone. Thus, if you quote something from Nelson, still provide the page number and the reference - for example: (Nelson 2016: p. 1697)

Question: How should we refer to examples from slides that weren't part of the curriculum.

Answer: You can refer to the slides, for example (Philosophy of Science, Lesson xx, slide xx).

Question: Is there a maximum number of characters we are allowed to write at the exam?

Answer: There is no formal limit. Generally longer answers get better grades, but that is not due to a causal effect of the length of the answer. Rather, longer answers tend to be better and length is thus just a confounding variable.

Question: Can you have your word documents in Onedrive, or does it have to be offline?

Answer: I don't think one needs to worry about any regular automated backup system that might be connected to the internet.

General comments (last updated 250322): I will answer some pollev.com/cabe questions in this Google Doc. I hope you understand that I will not be able to answer all questions posed. The more specific the question is, and the more votes it gets, the more attention I can give it.

I will "hide" questions if I think I've answered it, or if it was not clear or narrow enough to enable an answer. Feel free to try to make it more specific.

I've inserted some of the answers to questions from last year below.

You can cite or reference this Google Doc as: "Bergenholtz 2022: Google Doc Notes"

<https://docs.google.com/document/d/1mjmy196JRMc1XiFxPqI45a1mIAEDkzfBenV3iL3gJu8/edit?usp=sharing>

Answers to Pollev questions from this year 2022

I will add further answers in reverse order, so newest answers at the top - easier for you to see that something new has been added.

25.04.22

Question: Can you explain what is meant by "measurement and operationalization" being one of the 3 broad design problems in AMJ article? Specifically what is meant by operationalizing constructs? / Hvad menes der med "operationalizing constructs".

Answer: Primære budskab er at forskellige forskere kan have forskellige tilgange til at måle de begreber, de interesserer sig for. Virksomheds performance eller medarbejder motivation eller medarbejder performance kan måles på mange forskellige måder, dvs. både ud fra forskellige definitioner men også med forskellige tilgange til at måle. Måler man medarbejder performance ud fra lederens perspektiv, eller bruger man objektive data af en art. Eller, måler man performance hen over en længere periode eller blot en enkelt gang på et tidspunkt. Alle de her valg er udtryk for forskellige måder at operationalisere et construct på. In **English:** The primary message is that different researchers have different approaches to measure the concepts, they are interested in. Company performance or employee

motivation on employee performance can be measured in many different ways. Both taking departure in different definitions but also different approaches to measurements. Does one measure employee performance from the manager's perspective, or does one use objective data of sorts. Or, does one measure performance over a longer period, or just at one point in time. All these choices imply different operationalizations of a construct.

Question: Kan en teori nogensinde være fyldestgørende, hvis ikke der ikke er gjort brug af kvalitative metoder ifølge Sutton & Staw? De mener vel at man skal ind kvalitativt og undersøge årsagssammenhængene. / Can a theory ever be adequate if no qualitative methods are used?

Answer: Sutton & Staw argue that quantitative researchers should appreciate (=værd sætte) qualitative approaches more, but they are not arguing that the use of a qualitative method is a necessary criteria for obtaining (=opnå) a causal conclusion.

Question: Is p-hacking considered misconduct? / Er p-hacking (som omtalt i Aschwanden) udtryk for videnskabelig uredelighed?

Answer: That depends. Taking departure in Resnik & Stewart, it depends on the intention. Did the researcher know it was misconduct. Taking departure in the danish definition, it depends on if it constitutes gross negligence (=grov forsømmelighed). So, if the researcher clearly knew that it was p-hacking and it was wrong, then it could be misconduct (=videnskabelig uredelighed). In Carney's example, most would agree that this was not misconduct, because they did what most other researchers at the time also did. Only later did the field get wiser (=klogere).

Question: Should we include a list of references? and how?

Answer: You can just add it at the end of your exam.

Question: Kan du forklare ordet constructs? Hvilken artiklen fremgår det i? / What is a construct?

Answer: I suspect you have heard the term in quantitative methods. In any case, I've explained the concept in a previous year, so if you search for the word construct, you find an explanation (roughly page 7).

Question: In class you mentioned that if all physics books were burned we would still end up with those same books again after "redoing" the science but this would not happen in social sciences. I would like to be able to use this example but cant find out where it is from. / Hvor kommer eksemplet om "at brænde alle fysik bøger og resultater ville være det samme 1000 år senere" fra?

Answer: Ricky Gervais' video from lesson 1, about slide 17, see also <https://www.facebook.com/scienceteeshirts/videos/1915067131841719/>

Question: Når man vurdere rigor og relevant ift. Vermeulen og inddrager studier. Vil man så fx i felteksperimenter sige, at rigor er opfyldt, hvis fx RCT er opfyldt og er der andre kriterier man kan kigge på? / When you are assessing the rigour and relevance of a study, would you (e.g. in relation to fieldexperiments) say that rigour is fulfilled if it is an actual RCT?

Answer: Yes, if it is an actual RCT, it would be a rigorous study.

24.04.22

Question: Kan du forklarer hvordan vi vurderer om RCT er til stede i et forsøg/eksperiment? Hvilke trin skal man igennem? / Can you explain how to assess if a study is a RCT, what steps to go through?

Answer: Identify the R C T elements. Thus, does the research design involve a randomization, that is have participants been randomly placed in different conditions? Furthermore, is it clear that a treatment has been applied, that is there should be only one difference between the conditions. Finally, does it seem clear that the environment is controlled, so that the situation is more or less the same for participants. In a field experiment that is not easy to ensure, but maybe steps have been taken to ensure this.

Question: Kan du give eksempler på hvornår det er et felt eksperiment, felt studie og quasi eksperiment. / Can you give examples of when it is a field experiment, field study or quasi experiment.

Answer: In this Google Doc (further below) I address quasi vs. field experiments. If you search for the word quasi in this document, you should find it. Note I have written about it several times, the most extensive example is on page 9-10 (currently).

Question: Den tilsendte referenceliste er lidt inkonsistent i dens opstilling og referencemetode - kan vi være sikre på, at andre censorer også 100% vil godtage den referenceliste + referencer i teksten, som du har opstillet til os? Eller bør vi rette den til ift. Harvard referencing method for at være sikre? / is it fine to use the provided reference list, since the style seems a bit inconsistent (APA vs. Harvard style e.g.)

Answer: Yes, you can be sure it is fine to use the provided reference list. Feel free to adjust the style! If you do, feel very welcome to send the updated list to me.

Question: Please specify what DAG is?

Answer: DAGs are mentioned on page 22 (and later) in Barrowman 2014, but we were only to read page 1-16. They are thus not part of our curriculum. If interested one can find good explanations on Youtube, but they are not part of this course.

Question: Er det i orden at bruge at inddrage artikler fra tidligere års pensum? Ser du hellere at man benytter artikler fra dette års pensum, selv hvis opgaven ikke specifikt siger, at man skal referere til pensum? / Can you use articles from prior years' curriculum? Is it better to use this years' curriculum?

Answer: Any prior or current curriculum is fine. No preferences.

Question: Hvad er forskellen på Barrowman og BRM's tilgang til RCT? / What is the difference between Barrowman and BRM's approach to RCT?

Answer: There is no difference as such. Different perspectives and focus on different elements, but no difference as in fundamental disagreement.

Question: Can you have your word documents in Onedrive, or does it have to be offline?

Answer: I don't think one needs to worry about any regular automated backup system that might be connected to the internet.

Question: Hvad betyder selection bias? / What does selection bias mean?

Answer: Selection bias means that the results you get in a study might be due to the particular individuals/firms that end up being part of your study. The study is biased to the way data has been selected. Let's say you study how entrepreneurs become successful, but if you only study successful entrepreneurs, the results you get might be biased in the sense that non-successful entrepreneurs might engage in the same behaviors, but just fail for other not yet identified reasons. One could also study if playing musical instruments when young, could lead people to live a longer life. If one samples old people to study this question, you might have a biased sample in the sense that you don't have data on all the people that have died in the meantime, since they were young. Your sample might be biased, and a way to unbiased would be to sample young people and then follow them until they are old.

See also slide 32 (HA students) or slide 31 (Bsc students), lesson 4.

23.03.22

Question: Barrowman 2014 kan du ridse hans hovedpointer op? / What are the key points in Barrowman 2014.

Answer: The key take-aways (see also beginning of lesson 2) are: That correlations does not necessarily imply causation; why and how confounders (other variables that interfere) might prevent (=forhindre) easily achieving a good understanding (cf. ice cream, sun, sunburn example, or the psychologist Gottman); Barrowman also explains how we shouldn't expect single causes, but multiple causes for events, making an analysis more complicated (a bit in line with Aschwanden 2010); Finally; Barrowman also explains why and how, given all the challenges outlined, a randomized controlled trial provides insight into causal effects that other designs have difficulties getting to.

Overall, Barrowman 2014 thus sets up the agenda for experiments and research designs, and the challenge of identifying clear causal and scientific explanations for phenomena we are interested in.

Question: Kan du forklare paradigmer igen? Jeg tror aldrig jeg helt fangede den del af pensum ift paradigmeskift osv / Can you explain paradigms again?

Answer: This video contains Kuhn's paradigms (and shifts):

<https://youtu.be/CHQ7sCvwAoY>

Question: I Kuhn postscriptet er der angivet årstal 1969 i starten af artiklen, men i pensum- og referencelisten står der 1962. Hvilket årstal er korrekt at referere til? / How to refer to Kuhn, since it seems to be written in 1969, but is recorded in the reference list as being from 1962.

Answer: I understand the confusion. Kuhn's book was originally written in 1962, while the post-script we read is written in 1969, and published in 1970. You can refer to Kuhn 1962, but any year is fine with regards to this exam, due to the confusion.

Question: Hvordan skal vi referere til eksempler fra slides som ikke har været en del af pensum? / How should we refer to examples from slides that weren't part of the curriculum.

Answer: You can refer to the slides, for example (Philosophy of Science, Lesson xx, slide xx).

Question: How do we reference Kuhn vs PoS? Isn't the latter just further explanations of Kuhn's thoughts?

Answer: Yes, the latter (PoS) is a further explanation of Kuhn's original document. In order to explain Kuhn's position, you can refer to the text you find the most appropriate (=passende) in the situation.

Question: If the randomization in an experiment is poor, is it then a quasi experiment or an RCT with limited validity? Hvis randomiseringen i et eksperiment ikke er så god; Er det så et kvasieksperiment eller et RCT med begrænset validitet?

Answer: This depends (=afhænger af) on the problem with the randomization. Let's say that in a study of how meaningfulness (or lack of meaning) impacts how many Lego Bionicles one builds, and one by accident (=ved et uheld) has put all Lego nerds into one condition, then the randomization is deeply problematic, and it is not a RCT. One might imagine a situation where the randomization worked for 95 out of 100 people, but the last 5 all ended up in a condition without being randomized -> then we have a situation with a randomization that is not perfect, but overall it probably still works. In contrast, we talk about something as a quasi-experiment if the treatment (e.g. at what time of the day you take a test in elementary school) is not randomized, but more or less as good as random.

Question: Linking the perspectives of Guba and Vermeulen. Do they disagree? / Er Guba og Vermeulen uenige?

Answer: Yes, as indicated via the Guba quote in the Vermeulen slideset (lesson 13), Guba argues there is a tradeoff between rigor and relevance. The more one strengthens (=styrker) rigour, the less relevance one ends up with. Guba argues that the more one, for example, controls the environment and does a lab-study, the more rigour there is in the study, but the less relevant it would be for a real-world context. In contrast, Vermeulen argues that there is no tradeoff.

Question: Can you explain the differences again, between the danish and american definitions of misconduct / Kan du forklare forskellen mellem den danske og amerikanske definition af videnskabelig uredelighed?

Answer: The primary difference, briefly put, is if misconduct is defined by having to demonstrate a dishonest intention, or if one just has to show gross negligence, that is that the researcher should have known this was a wrong thing to do.

Question: Du skriver i referencelisten "You can quote an article as listed in the curriculum overview, thus e.g. write AMJ 2011 as a reference to the article by Bono and McNamara." Hvis man laver in-tekst kildehenvisninger til et citat, ville du så ved BRM teksten bare skrive (BRM, side x)? / how to quote, e.g. BRM document.

Answer: Yes, writing BRM and page number is fine (BRM, p. xx). This clearly shows it is a citation, and BRM of course also has to be listed in the reference list.

A few questions concerning the exam, and referencing

Question: Is there a maximum number of characters we are allowed to write at the exam?

Answer: There is no formal limit. Generally longer answers get better grades, but that is not due to a causal effect of the length of the answer. Rather, longer answers tend to be better and length is thus just a confounding variable.

Question: How to reference slides?

Answer: If you are quoting (=citere), you should refer to lesson as well as slide number. You can also refer to a lesson (Lesson X) if you want to link to a specific point made for example. In the reference list you can simply write "Bergenholtz, Lesson X", for the purpose of this exam, where I know what lessons you are referring to.

Question: How to prepare for the exam. Read all articles again? write down definitions? Do you have a checklist for a good answer? / Hvordan at forberede sig til eksamen

Answer: I am answering this question now, since it would be a bit late to wait to lesson 15 with answering this. Below I have inserted some of the prior slides on this topic.

A few comments in addition to the information in the slides below. Some of the articles are fairly difficult, so having to read a text twice should not be considered unusual. Having some prepared notes (e.g. including key quotes) for some of the texts is probably helpful - you might check out exams from 2018-2020 and note that some topics often appear. Generally, you can also consider if there are texts that have been mentioned often throughout the course.

A checklist for a good answer I don't have as such. My best recommendation is to practice former exam questions (e.g. spend at least 15-30 minutes considering what you would write) and then compare with the uploaded good answers.

From lesson 1. See also additional slides on this topic at the end of the slideset.

Qualitative insights from two very successful 2020 students

- Some of the texts are difficult – pay attention to reading guides' advice on what to focus on
- Produce written notes on texts' key message (incl. quotations to use), to ensure actual understanding
- Answer former exam sets + polleverywhere questions
- Consider the exam: Produce overview of what material and notes might be applicable for various types of exam questions
- Watch videos **before** lecture (when relevant)

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From lesson 12

A few comments about the exam

- Suggestion for preparation
 - Read curriculum + reflect on exercises/questions in class + study previous exams
 - You'll be too busy to read the curriculum in the exam situation
 - Practice Research Design challenges + look at former exam questions and consider how you would approach them. This year's question will be similar in style
 - This course doesn't call for a template approach, as some other courses.
 - Not a course where one (only) learns a model or table or technique: critical thinking and reflection on what science and truth is about is needed

For illustration, parts of some 2022 exam questions:

- Please identify the research designs involved in the above description, and assess their strengths and weaknesses...
- Which of the following sentences is XXX unlikely to have authored? Please be specific in your explanation of why some of the sentences do not match...
- Is the overall message provided in XXX the most similar to elements in XXX, XXX or XXX? Please provide a detailed explanation for your choice.

Below I have inserted questions and answers provided 2021

Question: Cross-sectional vs. longitudinal

Answer: I just want to highlight a distinction I made when presenting the experimental survey on experiment aversion, in lesson 15. The setup could be considered a comparison

over time (longitudinal), because we could compare responses in time period 1 with time period 2. We could do that because we sampled from students who either had not attended the class yet, or had attended it. In this way we can longitudinally compare the potential change, before vs. after the class - even though we don't know if identical people responded. However, had we not known that the people we sample are very similar, we could not just make the same inference, and interpret it as longitudinal.

Of course, had we only looked at data from the first time period, it would be a cross-sectional setup.

Furthermore, if you want to collect data on sleep, and decide to measure sleep every day for a full week, but only use the average - then it is a cross-sectional study, because you are not interested in change over time. Let's say the week of sleep was after an operation, then you might be interested in how sleep changes after the operation, and maybe one would assume more sleep on the last day. Then one wouldn't use the average, but each individual time point, and the study becomes longitudinal.

Question: Hvad er forskellen på Epistemologi og metodologi? / **What is the difference between epistemology and methodology?**

Answer: Epistemology concerns the standard for what knowledge it is possible to achieve. Methodology concerns the methods one uses to get there. So, if epistemologically speaking we can approximate the truth but have to acknowledge that the inquirer interacts with what we inquire into, then triangulation is meaningful, thus applying a range of different methods (methodology).

Question: What is a construct?

Answer: A small extract from a textbook: "In the context of survey research, a construct is the abstract idea, underlying theme, or subject matter that one wishes to measure using survey questions. Some constructs are relatively simple (like political party affiliation) and can be measured using only one or a few questions, while other constructs are more complex (such as employee satisfaction) and may require a whole battery of questions to fully operationalize the construct to suit the end user's needs. Complex constructs contain multiple dimensions or facets that are bound together by some commonality that, as a whole, compose the construct." <https://methods.sagepub.com/reference/encyclopedia-of-survey-research-methods/n91.xml>

A construct is thus a word we use for measuring the phenomena we are interested in (innovation, unemployment, motivation etc. are all constructs), and one can have different constructs (ways of measuring something) for similar phenomena.

Question: Heisenberg uncertainty principle - not mentioned in the curriculum, right?

Answer: I've added Heisenberg's uncertainty point to illustrate the challenge of investigating the world, even in (quantum) physics. It is not curriculum as such, but use it if it helps you make sense of something.

Reference list II

Question: How to refer to tweets, statements, videos etc. from teaching

Answer: It is fine to refer to the slides where this is mentioned. It is also ok to more generally state "as Ricky Gervais says..." or "as the Heisenberg quote states"...If you are quoting the tweet, then you of course have to cite the slide.

Question: Slides fra studiecafé 1 lister regler til at vurdere kvaliteten af et forskningsdesign. Vurderes kvaliteten forskelligt afhængigt af, om det er et eksperiment eller ej (siden en slide kun omhandler eksperimenter, og en anden er generel)? **Studycafe 1 lists rules for assessing the quality of a research design. Is the quality assessed differently, depending on if it is an experiment or not (one slide is about experiment, the other one is general)?**

Answer: A fair question that I have also received in the question hours a couple of times. Assessing an experiment can follow a more well-defined recipe, since they are more standardized and have to follow clear criteria. In contrast, research designs in general (cross-sectional, longitudinal, cases) are much more heterogeneous and as clear guidelines can not be established. Nevertheless, the general principles of the guidelines for experiments can be transferred. E.g., a challenge to external validity of a RCT is if there is an interaction between history and treatment - similarly, if the data that a given cross-sectional study (or any other) relies on is old, then that can also limit external validity (e.g. if it is about use of technology, just as an example). Similarly, what sample (selection) is in play, can also matter. So, again, the principles can be transferred, but guidelines are not as well specified.

Question: Hvad er koblingen mellem Resnik & Stewart og Kuhn? Kan man lave en lighed mellem Resnik & Stewart ift. normer som ligger inden for scientific practice med Kuhns fortolkning af 'normal science'? Og dernæst, når Resnik & Stewart taler om noget som falder uden for normerne ift. scientific practice... / **What is the connection between Resnik & Stewart and Kuhn?**

Answer: Yes, as highlighted early in lesson 11, Resnik & Stewart are arguing that in Kuhn's "normal science" it is easier to agree on relevant norms for scientific practice, and thus also what dishonest science then might be. In contrast, if there are paradigmatic and fundamental disagreements across different communities, then the "right" norms can be more difficult to assess.

Question: Vil du kort, igen, forklare hvad rigour og relevant er? / **Can you, briefly, again explain what rigour and relevance is?**

Answer: The best I can do is to refer back to the video from lesson 13. The question needs to be more specific, to enable me to add further perspectives. From about 8.40

<https://youtu.be/0qgg6xjBC6c>

Question: Vil felt eksperimenter oftere være QUASI på grund af manglende RCT? **Would field experiments often be quasi due to lack of RCT?**

Answer: No, if one has randomly allocated different people (workgroups, units etc.) to different conditions, then it is a RCT. We would often have to acknowledge that it is more difficult to control an experiment that does not take place in a lab. So, field experiments are still proper RCTs, but with less certainty about maintaining 100% control. It would become quasi if the allocation to conditions was not entirely random.

Question: Udover selvfølgelig at læse teksterne, hvad vil dine bedste eksamensforberedelses-tips være? At finde citater fra diverse tekster? At lave små opsummeringer af dem? Noget tredje? / **Besides reading the texts, what is the best way to prepare for the exam? Finding quotes from various texts? Making summaries, something else?**

Answer: In lesson 1 (at the end) I have provided some general reflections on how to prepare for an exam, and also conveyed insights from former (excellent) students. As the general reflections convey, just re-reading is not particularly efficient. Practicing former exam questions (and comparing to good answers), making notes and finding quotes sounds like a good idea.

Question: 2019 exam question 4 refers to Hambrick and Sutton & Staw. Is this curriculum?

Answer: No, I have now specified that these papers are not included in this year's curriculum, and of course not something you are expected to know anything about.

Question: Hvordan vurderer man den interne og eksterne validitet for andre forskningsdesign end eksperimenter (cross-sectional, longitudinal og case studier)? Kan man bruge de udfordringer, der er nævnt i AMJ 2011 til at begrunde, hvorfor der kan være problemer med intern/ekstern validitet generelt? / How do you assess the internal and external validity of other research designs than experiments (cross-sectional, longitudinal and case studies)? Can you use the challenges that are mentioned in AMJ 2011 to explain, why there can be problems with the internal and external validity?

Answer: Yes, exactly. Question 5 in the 2018 exam (Brenda) is a good illustration of this. As AMJ 2011 highlights (and Barrowman 2014), it is difficult to make a causal claim based on cross-sectional research, since you only have insight on variables at the same point in time; what came first, and what other variables might explain the outcome? Furthermore, AMJ 2011 also talks about selection bias - as in the WW2 airplane or 'only studying successful entrepreneurs' examples. AMJ 2011 also discusses the quality of a construct; how well have we measured what we are interested in? In the Brenda example the researcher has multiple approaches to measuring leadership and employee performance. Also think of the Danish facemask study: they said they had studied the ability of facemasks to stop one from getting Covid-19, but actually they had compared one group that received 50 masks and an instructional video vs. a group that did not receive masks. The AMJ 2011 paper - and the video + slides, present a few more examples.

Question: Hvad kræves der, for at man har med et kvasi-eksperiment at gøre? / What constitutes a quasi-experiment?

This question was also addressed at the end of lesson 13 (live session), last 3 slides.

Answer: RCTs have a clear definition: It requires a randomization allocation of participants into different groups, only changing one variable in a controlled environment. Only one feature is different in the treatment group vs. the control group.

However, quasi-experiments are trickier to define. A quasi experiment does not entail an actual randomization by scientists, but still allows for a comparison across groups where (hopefully, and basically) only one variable is the difference. A few [examples](#) from class and then some [criteria](#) just below the examples.

[Example 1](#), Cognitive fatigue: We have heard about the elementary school study, investigating the influence of cognitive fatigue on test performance; the later in the day, the worse the performance. No formal randomization process is implemented. Yet, kids do not themselves decide when to take the test and the ones who allocated classes to different time slots do not do so in a way that is systematically linked to the abilities of kids - which

otherwise would skew the results. Overall, we can assume that there are no systematic differences between those kids that take a test early vs. late. However, if we for some reason find out that many schools decide to systematically vary when certain kinds of kids take a test, we would stop categorizing this as a quasi-experiment.

Example 2, Spanish flu: This example (lesson 13 among others) did not constitute a quasi-experiment, since the cities could self-select into implementing non-pharmaceutical interventions themselves. So, it would be like kids deciding for themselves when to take the test. Furthermore, the kids also engaged in all sorts of other decisions that could influence mortality - e.g. making masks mandate, communicating effectively about how not to infect others etc. Hence, the only difference between cities is not how early they closed down. In order for something to be a quasi-experiment we want to be fairly certain that there is only one difference between the people, organisations, countries, cities (etc.) in the experiment. So, **criteria:** A quasi-experiment does not require a randomization implemented by researchers. However, we still need to i) compare two groups, that ii) are exposed to different 'treatments', and iii) the people in the different groups can't have self-selected into their group, but should be 'allocated' based on a process that is as good as random. For example, we can't compare people that start a business with those that don't and say; look, the only difference is if they started a company – because the people would have made the choice themselves, meaning there is nothing random about it.

Natural experiments: I will not refer to natural experiments in exams.

The radioactive rainfall example is a natural experiment. See Wiki for a definition: "A natural experiment is an empirical study in which individuals (or clusters of individuals) are exposed to the experimental and control conditions that are determined by nature or by other factors outside the control of the investigators", https://en.wikipedia.org/wiki/Natural_experiment. The cognitive fatigue study would usually not be categorized a natural experiment, since humans allocated test times.

Reference list I

Question: Er det et krav, at man skriver en litteraturliste til sidst i sin eksamensopgave? / **Is it a requirement to add a reference list at the end of your exam answer?**

Answer: Yes. I have added a complete reference list on Blackboard (folder 'Reference list') that you can use, in order to save time. You only have to add those you relied upon, of course. Due to the time-limited nature of the exam, it is fine that you have a list that is too long. In general, reference lists should only include papers you actually used.

Research designs I

Cross-sectional vs. longitudinal

A good question in the Zoom session encouraged me to emphasize the following point. A cross-sectional study becomes longitudinal when there are multiple data points, but of course these need to actually be on a relevant timeline. In other words, collecting data twice on sleep (as in the 2019 exam), but without collecting data on what comes first or even using the order of the first and second data point, then it doesn't constitute a longitudinal element. As the examples in the video illustrates, longitudinal data involves that you collect data at one point in time, and then follow up later, to see what change there has been. E.g. video games played at year 12 and 13, to see how aggressive behavior might have changed from one year to the next, depending on the amount of games played in the meantime. So, one needs multiple data points, but these need to be over time, and they need to show how something changed from one point in time, to the next, as illustrated in the video.

Question: Forklaring på casestudies og hvordan de adskiller sig fra long/cross / Explanation of case studies and how they are different from longitudinal / cross-sectional studies

In a **case study** they aim is to understand the particular context, that is to collect rich data on the case. The case can be defined as a classroom, a company, a group of companies, one individual, all customers in a store at a given time etc. It is a bounded (afgrænset) event, situation or organisation.

Thus, if one wants to understand the relation between playing violent video games and aggressive behavior, one could e.g. interview kids, their parents and friends in order to understand and explain the potential relation (e.g. take departure in all kids in a classroom, or in a team of computer players). But one would focus on the situation of each individual; why does Peter do what he does. Inductively one might be able to find patterns across different individuals in one's case study - or deductively test a theory.

In a **cross-sectional study**, one might also want to understand the relation between playing violent video games and aggressive behavior, but one is aiming to compare, e.g. across kids. One is not interested in one particular situation, but the average relations across a bunch of cases. One therefore has a (well-defined) sample and characterizing the sample is the aim. With a case, the aim is characterizing each individual context: What does Peter do? In a cross-sectional approach, the question would be: What do young males do? (e.g).

A **case study** can become **longitudinal** if you follow these kids over time. Not just over a few hours, but over enough time allowing for a comparison between time 1 and time 2.

Similar with **cross-sectional**: A study goes from being cross-sectional to **longitudinal**, when you have multiple data points on the same individual (or firm or country etc.), with the aim of making comparisons across time, comparing the same individual at multiple time points.

Note that it doesn't help if one has data on variable X in time point 1 and variable Y in time point 2 - this does not enable a direct comparison over time, of how a certain variable (e.g. performance) changes over time.

Answers to 2020 questions

Page numbers

Sometimes page numbers differ, depending on which version of the article one has downloaded. Same applies to the BRM bookchapters that have 2 sets of page numbers. Don't worry about that issue.

On reducing uncertainty, and causality

I'd like to repeat a point I made in the Lesson 2 Correlation vs Causation video:

https://www.youtube.com/watch?v=Wz-1b_B6lnU

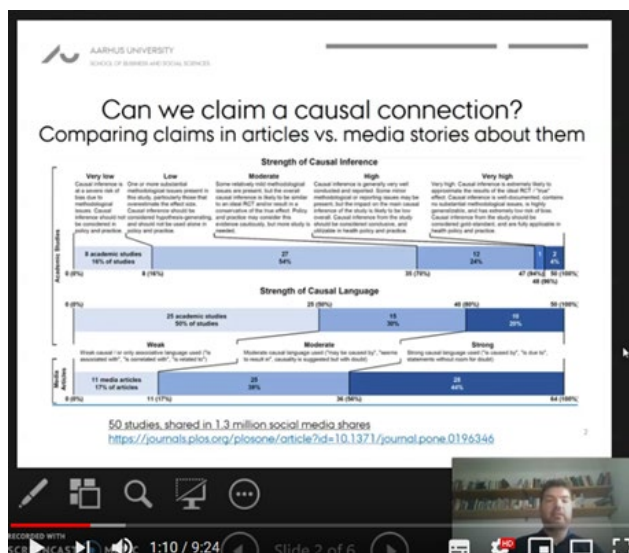
In the class we have focused quite a bit on experimental designs, and how one can't claim a causal effect from a cross-sectional setup or a longitudinal design that doesn't take selection effects into account (e.g. the private vs. public school ex. from Study Cafe 2). However, as

argued in the Causality video, in real life we often have to make decisions with less than certain causality information. As the Aschwanden (2015) text and the overview early in the Causality video shows, it is about reducing uncertainty. A big cross-sectional study is better than a small one. Not merely asking a small kid if (s)he is aggressive and/or plays violent video games, but actually obtaining an external assessment of the games' violence as well as the kid's aggressive behavior, is better. Engaging in a longitudinal study to try to capture development over time is even better. Trying to address the selection bias in the private-public school ex. is better than ignoring it – e.g. by considering the kids cognitive ability before they start school, when trying to figure out which kind of school improves ability the most. Of course, we would like to see a field experiment, yet that is often not possible.

So, cf. the 2019 re-exam video (https://www.youtube.com/watch?v=kH_BXsAmEt8, beginning of the video), when we have data comparing two cities that appear to identify a key variable (social distancing) - we should go with this evidence, the current best one, until better information comes along. We often can't wait for the perfect randomized controlled trial.

This tweet sums it up, in its own way:

<https://twitter.com/petergklein/status/1239642933617397765?s=20>. In other words, I have focused on the difficulty of identifying actual causality, in order to make us more skeptical about claims to causality we often see. This also makes a general point about what science can do, and how difficult it is to get to the Truth. However, we should also accept that science is about reducing uncertainty (Aschwanden 2015), as my use of the video games example tried to illustrate, and as the visual overview in the Causality video shows (see below). If we can move further to the right in the overview below, that's better than making uninformed correlational guesses.



Kuhn

Just to be clear: A new paradigm might be 'evolutionary' better, but it will not be a more true description of reality. There is no 'theory-independent' perspective, that would warrant such a claim.

Guba 1990

The ism 'Critical theory' is not part of our curriculum.

Lab vs. quasi experiments vs. natural experiments

- A lab-experiment should not be a quasi experiment. The point of a lab-experiment is that one CAN control and randomize. Hence, if it is in effect a quasi-setup, it is most likely a failure.
- Natural vs. quasi experiment: I have focused on the difference between a quasi and RTCs -this is the key difference you should be able to identify. A natural experiment is defined by something happening that was outside the researchers control. The elementary school (cognitive fatigue) example can therefore be characterized as a natural experiment - but since the allocation wasn't 100% random, it is also a quasi experiment. A quasi experiment could be organized by a researcher who does have some access to a firm, but can't quite implement a proper randomized control. There is also slight disagreement on differences between natural and quasi experiments, which is why I have not focused on this. Understanding what an RTC is, and when the criteria are not quite fulfilled, and what that means - that is important for this exam, and understanding research in general.

Vermeulen: On relevance

An additional comment on Vermeulen's take on relevance. As is made clear in his text, he focuses on the question. He could have said it should be a matter of external validity, e.g., just as he insists on strong internal validity (rigour). His approach is not to do that, but to encourage researchers to ask relevant questions that can generate insights. A lab-study can generate insights, although it of course requires some translation by the practitioner. But again, Vermeulen did not say that the question (or answer) should solve immediate problems in the organisation.

Yet, if the research question is about a concept that a manager can't do anything about (the weather, to provide a crude example) e.g., or based on outdated data, or is about an issue that is not important for a manager - then the question can't be relevant.

Being an online student

Something more general. Part of this course is now online, and you are - and in the future will be - part of more online courses. This Twitter thread has posted some really good advice for students: how to learn in an online environment:

<https://twitter.com/katesymons2/status/1239490473556877312>